EE2023 Final Exam Cheat Sheet

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1. Signals: Energy & Power

Total Energy E: $E = \int_{-\infty}^{\infty} |x(t)|^2 dt$.

Average Power P: $P = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |x(t)|^2 dt$.

For periodic signals: $P = \frac{E_{\text{one period}}}{T} = \frac{1}{T}\int_0^T |x(t)|^2 dt$.

RMS Value: $X_{rms} = \sqrt{P}$.

Energy/Power Classifications:

- **Energy Signal:** $0 < E < \infty \implies P = 0$ (transients, finite duration).
- **Power Signal:** $0 < P < \infty \implies E = \infty$ (periodic, step, constants).
- **Neither:** $E = \infty$ and $P = \infty$ (e.g., $t \cdot u(t)$).

Transformations on Energy & Power: If $y(t) = A \cdot x(at - b)$, then:

$$E_y = \frac{|A|^2}{|a|} E_x, \quad P_y = |A|^2 P_x$$

Note: Time shifting (b) and reversal do not affect E or P. Compress ($a > 1$) reduces energy. Expand ($a < 1$) increases energy. Power is unaffected by time scaling.

Common Energies: - $\text{rect}(t/T) \implies E = T$ - $\text{tri}(t/T) \implies E = \frac{2T}{3}$

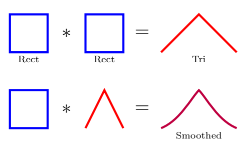
- $A\,\text{sinc}(Wt) \implies E = \frac{A^2}{W}$ (via Parseval's area of squared rect).

2. LTI Systems & Convolution

Convolution Integral: $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

Graphical Convolution Steps: 1. **Fold:** Flip $h(\tau)$ to get $h(-\tau)$. 2. **Shift:** Slide by t to get $h(t-\tau)$. 3. **Multiply:** Multiply $x(\tau)$ and $h(t-\tau)$ for a specific t. 4. **Integrate:** Area under the product is $y(t)$.

Common Shapes (Graphical Intuition):



Rect * Rect Shortcut & Equations:

- **Equal Width (T):** $A\,\text{rect}(\frac{t}{T}) * B\,\text{rect}(\frac{t}{T}) = ABT\,\text{tri}(\frac{t}{T})$.

Piecewise Eq: $y(t) = ABT(1 - \frac{|t|}{T})$ for $|t| \le T$, 0 elsewhere.

- **Unequal Width ($T_1 > T_2$):** Yields a Trapezoid.

Max Height: ABT_2. *Total Base Width:* $T_1 + T_2$.

Flat Top Width: $T_1 - T_2$ (flat from $t = -\frac{T_1-T_2}{2}$ to $+\frac{T_1-T_2}{2}$).

Properties of Convolution:

- **Commutative/Distributive/Associative** rules apply.
- **Derivative:** $\frac{d}{dt}[x(t) * h(t)] = \frac{dx(t)}{dt} * h(t) = x(t) * \frac{dh(t)}{dt}$
- **Integration:** $\int[x(t) * h(t)]dt = [\int x(t)dt] * h(t)$
- **Impulse/Shift:** $x(t) * \delta(t - t_0) = x(t - t_0)$
- **Width Rule:** If $x(t)$ has length L_1 and $h(t)$ has length L_2, $y(t)$ has length $L_1 + L_2$. Start time = sum of start times.

LTI System Properties (from Impulse Response $h(t)$):

- **Memoryless:** $h(t) = K\delta(t)$.
- **Causal:** $h(t) = 0$ for $t < 0$. (Output depends only on past/present inputs).
- **Stable (BIBO):** $\int_{-\infty}^{\infty} |h(t)|dt < \infty$ (Absolutely integrable).
- **Invertible:** There exists $h_{inv}(t)$ such that $h(t) * h_{inv}(t) = \delta(t)$.

Step Response $s(t)$: $s(t) = \int_{-\infty}^{t} h(\tau)d\tau = h(t) * u(t)$.

Inverse relationship: $h(t) = \frac{ds(t)}{dt}$.

3. Fourier Series (Periodic Signals)

For a periodic signal $x_p(t)$ with period T_p and fundamental freq $f_p = 1/T_p$: **Complex Exponential FS:**

$$x_p(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_p t} \quad | \quad c_k = \frac{1}{T_p}\int_{-T_p/2}^{T_p/2} x_p(t)e^{-j2\pi k f_p t}dt$$

Trigonometric FS Form:

$$x_p(t) = a_0 + \sum_{k=1}^{\infty}[a_k\cos(2\pi k f_p t) + b_k \sin(2\pi k f_p t)]$$

Relation to c_k: $c_0 = a_0$, $c_k = \frac{1}{2}(a_k - jb_k)$, $c_{-k} = \frac{1}{2}(a_k + jb_k)$.

Symmetry Properties: - Real $x(t) \implies c_{-k} = c_k^*$, $|c_k|$ is even, $\angle c_k$ is odd. - Real & Even $x(t) \implies c_k$ is real and even ($b_k = 0$). - Real & Odd $x(t) \implies c_k$ is purely imaginary and odd ($a_k = 0$).

- **Half-Wave Symmetry:** $x(t) = -x(t \pm T_p/2) \implies c_k = 0$ for all EVEN k. Only odd harmonics exist.

DC Component: $c_0 = \frac{1}{T_p}\int_{T_p} x(t)dt$ (Average value over one period).

Parseval's for FS: Total Power $P = \sum_{k=-\infty}^{\infty} |c_k|^2 = a_0^2 + \frac{1}{2}\sum(a_k^2 + b_k^2)$.

Gibbs Phenomenon: Overshoot of $\approx 9\%$ at discontinuities when reconstructing a signal with truncated FS series.

4. FT Tricks & Spectral Density

Duality Trick: If $x(t) \rightleftharpoons X(f)$, then $X(t) \rightleftharpoons x(-f)$.

Example: We know $\text{rect}(t) \rightleftharpoons \text{sinc}(f)$. By duality, $\text{sinc}(t) \rightleftharpoons \text{rect}(-f) = \text{rect}(f)$ (since rect is even).

Energy Spectral Density (ESD): $E_x(f) = |X(f)|^2$ (Joules/Hz).

Total Energy $E = \int_{-\infty}^{\infty} E_x(f)df$ (Rayleigh Energy Theorem).

Power Spectral Density (PSD): For periodic signals, $P_x(f) = \sum_{k=-\infty}^{\infty} |c_k|^2\delta(f - kf_p)$ (Watts/Hz).

Total Power $P = \int_{-\infty}^{\infty} P_x(f)df = \sum |c_k|^2$.

Transmission through LTI Systems:

Let $y(t) = x(t) * h(t) \implies Y(f) = X(f)H(f)$.

- **ESD relation:** $E_y(f) = |H(f)|^2 E_x(f)$.
- **PSD relation:** $P_y(f) = |H(f)|^2 P_x(f)$.

Autocorrelation Function $R_x(\tau)$: For energy signals: $R_x(\tau) = \int x(t)x(t+\tau)dt \rightleftharpoons E_x(f)$.

For power signals: $R_x(\tau) = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} x(t)x(t+\tau)dt \rightleftharpoons P_x(f)$.

Note: $R_x(0) = E$ (energy signals) or P (power signals).

Bandwidth Definitions:

- **Half-Power (3-dB) BW:** Freq where $|X(f)| \ge \frac{1}{\sqrt{2}}|X(f)_{max}|$.
- **First-Null BW:** Freq where main lobe of $|X(f)|$ first hits zero.
- **Modulated BW:** When a lowpass signal is modulated to a high freq, Passband BW = $2\times$ Baseband BW.

5. Amplitude Modulation (AM)

DSB-SC (Double Sideband Suppressed Carrier):

$y(t) = x(t)\cos(2\pi f_c t)$.

Spectrum: $Y(f) = \frac{1}{2}[X(f - f_c) + X(f + f_c)]$.

Demodulation (Coherent): Multiply received $y(t)$ by local oscillator $2\cos(2\pi f_c t)$, yielding $x(t) + x(t)\cos(4\pi f_c t)$. Pass through a Low-Pass Filter (cutoff $f_{max} < f_{cutoff} < 2f_c - f_{max}$) to recover $x(t)$.

Standard AM (with Carrier):

$y(t) = A_c[1 + \mu x(t)]\cos(2\pi f_c t)$, where μ is modulation index. Requires $\mu \le 1$ to avoid overmodulation (envelope distortion). Can be demodulated cheaply using an **Envelope Detector**.

Power in Standard AM:

Total Power $P_t = P_c\left(1 + \frac{\mu^2 P_x}{A_c^2}\right)$. If $x(t)$ is a normalized sine wave, $P_t = P_c(1 + \frac{\mu^2}{2})$ where $P_c = \frac{A_c^2}{2}$.

Power Efficiency η: $\eta = \frac{\text{Sideband Power}}{\text{Total Power}} = \frac{\mu^2}{2+\mu^2}$. Maximum efficiency is strictly 33.3% at $\mu = 1$.

6. Laplace Transform & ROC

Transfer Function: $H(s) = \frac{Y(s)}{X(s)} = \mathcal{L}\{h(t)\}$ (Zero initial cond).

Region of Convergence (ROC): Values for s where integral converges.

- **Right-Sided:** (e.g., $e^{-at}u(t)$) $\implies$ ROC is $\text{Re}\{s\} >$ Max Pole.
- **Left-Sided:** (e.g., $-e^{-at}u(-t)$) $\implies$ ROC is $\text{Re}\{s\} <$ Min Pole.
- **Two-Sided:** ROC is a vertical strip between two poles.

Initial Value Theorem (IVT): $\lim_{t\to 0^+} x(t) = \lim_{s\to\infty} sX(s)$.

Final Value Theorem (FVT): $\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s)$.

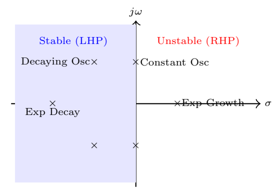
CRITICAL FVT RULE: Roots of the denominator of $sX(s)$ MUST all have negative real parts. If poles are on $j\omega$ axis or RHP, limit oscillates or is ∞.

Solving ODEs with Initial Conditions:

$\mathcal{L}\{\frac{dy(t)}{dt}\} = sY(s) - y(0^-) \quad \mathcal{L}\{\frac{d^2y(t)}{dt^2}\} = s^2Y(s) - sy(0^-) - y'(0^-)$

7. Pole-Zero Maps & System Behavior

Roots of the numerator are **Zeros** (O). Roots of denominator are **Poles** (X).



Stability: Stable LTI system $\iff$ ROC includes the $j\omega$ axis ($\text{Re}\{s\} = 0$). For causal systems, **all poles** must lie in open LHP ($\text{Re}\{s\} < 0$).

Dominant Poles: Poles closest to the $j\omega$ axis dominate the transient response because their exponential decay is the slowest.

Damping Ratio ζ for Complex Poles $s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$: - $\zeta = 0$: Undamped (Oscillates forever at ω_n, poles on $j\omega$ axis). - $0 < \zeta < 1$: Underdamped (Decaying oscillations, complex LHP poles). - $\zeta \ge 1$: Critically/Overdamped (No oscillation, real LHP poles).

Pole Categories & System Responses:

Category 1: LHP ($\sigma < 0$) $\implies$ Strictly BIBO Stable • *1. Strictly Real (e.g., $s = -3$):* $h(t) = e^{-3t}u(t)$. Non-oscillatory. Smoothly and exponentially decays to zero (like a weight in thick honey).

• *2. Complex Conjugate (e.g., $s = -1 \pm j3$):* $h(t) = e^{-t}\cos(3t)u(t)$. Oscillatory (Underdamped). Bounces decay to zero (like a car suspension).

• *3. Repeated Real (e.g., double $s = -3$):* $h(t) = te^{-3t}u(t)$. Non-oscillatory (Critically Damped). Rises slightly, then decays to zero.

Category 2: RHP ($\sigma > 0$) $\implies$ Strictly Unstable • *4. Strictly Real (e.g., $s = +3$):* $h(t) = e^{3t}u(t)$. Non-oscillatory. Grows exponentially toward infinity (like mic feedback).

• *5. Complex Conjugate (e.g., $s = +1 \pm j3$):* $h(t) = e^{t}\cos(3t)u(t)$. Oscillatory (Growing). Bounces grow larger until destruction.

Category 3: Imaginary Axis ($\sigma = 0$) $\implies$ Marginally Stable • *6. Exactly at Origin ($s = 0$):* $h(t) = e^0 = 1 \cdot u(t)$. Non-oscillatory. Pure Integrator, holds a constant value forever without decaying.

• *7. Complex Conjugate (e.g., $s = \pm j3$):* $h(t) = \cos(3t)u(t)$. Oscillatory (Undamped). Pure eternal sine wave, never losing/gaining energy.

• *8. Repeated on $j\omega$ (e.g., double $s = \pm j3$):* $h(t) = t\cos(3t)u(t)$. Oscillatory (Growing). Unstable, bounds grow linearly to ∞.

System Analysis Examples: • **System A (Poles at $-1, -3, -5$):** All poles are strictly on the real axis with no imaginary components. The impulse response is a sum of decaying exponentials ($e^{-t} + e^{-3t} + e^{-5t}$).

Oscillation: None. **Stability:** Stable (all LHP). • **System B (Poles at $-3, -1 \pm j3$):** The real parts are strictly negative. The complex conjugate pair introduces sine/cosine terms. **Oscillation:** Yes, oscillates within an exponentially decaying envelope ($e^{-t}\cos(3t)$). **Stability:** Stable (all LHP).

8. System Modeling & Feedback

Block Diagram Reduction Rules: - **Series:** $G_{eq}(s) = G_1(s) \cdot G_2(s)$ - **Parallel:** $G_{eq}(s) = G_1(s) + G_2(s)$ - **Feedback Loop:** $H_{CL}(s) = \frac{G_{forward}}{1 \mp G_{forward} H_{feedback}}$ (use + for negative feedback, - for positive feedback).
Characteristic Equation for stability: $1 \pm G(s)H(s) = 0$.
Steady-State Error (e_{ss}): For a unity negative feedback system ($H(s) = 1$), the Error signal is $E(s) = X(s) - Y(s) = X(s)\frac{1}{1+G(s)}$.
Apply FVT: $e_{ss} = \lim_{s\to 0} sE(s)$.
- **Step input** ($X(s) = 1/s$) $\implies e_{ss} = \frac{1}{1+K_p}$ where $K_p = \lim_{s\to 0} G(s)$.
- **Ramp input** ($X(s) = 1/s^2$) $\implies e_{ss} = \frac{1}{K_v}$ where $K_v = \lim_{s\to 0} sG(s)$.

9. Frequency Response & Filters

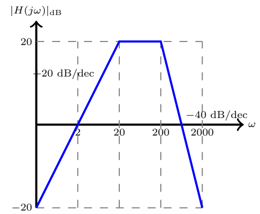
Substitute $s = j2\pi f$ (or $s = j\omega$) into $H(s)$ to find Frequency Response $H(f)$.
Sinusoidal Steady-State Response: If input is $x(t) = A\cos(2\pi f_0 t + \theta)$, output is:
$y_{ss}(t) = A|H(f_0)|\cos(2\pi f_0 t + \theta + \angle H(f_0))$
Golden Rule: Frequency ω_0 never changes; only Mag and Phase do.
Sine/Cosine Swap: $\sin(\theta) = \cos(\theta - 90^\circ)$ or $\cos(\theta - \frac{\pi}{2})$ **Fix Negative Amplitude:** $-A\cos(\theta) = A\cos(\theta \pm 180^\circ)$ **Combined Input Tip:** For $x(t) = C\cos(\theta) + S\sin(\theta)$, use: $\sqrt{C^2+S^2}\cos(\theta - \tan^{-1}(S/C))$
Distortionless Transmission:
Output is a purely scaled and delayed input: $y(t) = Kx(t - t_d)$. Implies: $H(f) = Ke^{-j2\pi f t_d}$. - **Magnitude:** $|H(f)| = K$ (Constant gain). - **Phase:** $\angle H(f) = -2\pi f t_d$ (Linear phase through origin).
- **Group Delay:** $\tau_g(f) = -\frac{1}{2\pi}\frac{d}{df}\angle H(f) = t_d$ (Constant).
Butterworth Low-Pass Filter: Maximally flat passband.
Magnitude: $|H(f)| = \frac{1}{\sqrt{1+(f/f_c)^{2n}}}$, where n is filter order.
At $f = f_c$, gain is $1/\sqrt{2}$ (-3 dB). Roll-off is $-20n$ dB/decade.

10. Bode Plots Construction & ID

Graphical representation of Mag $20\log_{10}|H(f)|$ dB vs $\log_{10}(f)$, and Phase $\angle H(f)$ vs $\log_{10}(f)$.
Summary Rule of Thumb for Asymptotes: • **At VERY HIGH freq ($s \to \infty$):** Only term with highest power of s matters. • **At VERY LOW freq ($s \to 0$):** Only term with lowest power of s (usually constant) matters. Tiny fractions shrink s to zero.
Asymptote Behavior Rules: • **Flat Line:** Horizontal line means constant gain (a regular number, K). • **Bend UP (+20 dB/dec):** Caused by a Zero (Numerator) at that specific corner frequency. • **Bend DOWN (-20 dB/dec):** Caused by a Pole (Denominator) at that specific corner frequency. • **Double Bend (±40 dB/dec):** Caused by a Double Pole or Double Zero (a term squared).
Integrators (K/s): Dictating Start Position The term (e.g., $10/s$) dictates your starting slope and anchor point. • **The Slope:** Any lone s in the denominator means you start with a downward slope of -20 dB/dec. • **The Anchor Point:** Find exactly where this line lives by plugging $\omega = 1$ into just this term. Gain at $\omega = 1 \implies |10/j1| = 10$. In dB: $20\log_{10}(10) = 20$ dB. Put a dot at ($\omega = 1, y = 20$ dB). Go one decade right ($\omega = 10$) and drop 20 dB ($y = 0$ dB). Draw line.
Simple Zero: $1 + j\frac{\omega}{\omega_c}$
- **Mag:** 0 dB for $\omega < \omega_c$, +20 dB/dec slope for $\omega > \omega_c$. Exact deviation: +3dB at ω_c, +1dB at $0.5\omega_c$ and $2\omega_c$.
- **Phase:** 0° for $\omega \le 0.1\omega_c$, $+90^\circ$ for $\omega \ge 10\omega_c$. Straight line $+45^\circ$/dec in between. Exactly $+45^\circ$ at ω_c.
Simple Pole: $\frac{1}{1+j\frac{\omega}{\omega_c}}$
- **Mag:** 0 dB for $\omega < \omega_c$, -20 dB/dec slope for $\omega > \omega_c$. Exact deviation: -3dB at ω_c, -1dB at $0.5\omega_c$ and $2\omega_c$.
- **Phase:** 0° for $\omega \le 0.1\omega_c$, -90° for $\omega \ge 10\omega_c$. Straight line -45°/dec in between. Exactly -45° at ω_c.
Second-Order System (Complex Conjugates):

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- **Mag Slope:** 0 dB at low freq, -40 dB/dec slope at high freq starting at corner ω_n.
- **Phase:** Drops from $0^\circ \to -180^\circ$ over 2 decades centered at ω_n, passing exactly through -90° at ω_n.
- **Resonance Peak:** Exists only if $\zeta < \frac{1}{\sqrt{2}} \approx 0.707$.
- **Peak Frequency:** $\omega_r = \omega_n\sqrt{1 - 2\zeta^2}$.
- **Peak Magnitude:** $M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$. To convert to dB use $20\log_{10}(M_r)$.
System Identification from Bode Plot: 1. **Initial Slope:** Slope at $\omega \to 0$ gives poles/zeros at origin. (e.g. -20N dB/dec = N poles at origin). 2. **Corner Frequencies:** Changes in slope indicate poles (slope decreases by 20) or zeros (slope increases by 20).



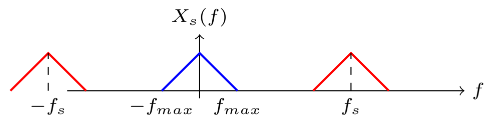
Example (Reconstruct $H(s)$): Mag plot rises at +20dB/dec, crosses 0dB at $\omega = 2$. Flattens to 0 slope at $\omega = 20$. Drops to -40dB/dec at $\omega = 200$. • **Origin:** Initial +20dB/dec means an origin zero Ks. $20\log(K \cdot 2) = 0 \implies K = 0.5$. Term: $0.5s$. • **Corners:** At $\omega = 20$, slope changes $+20 \to 0$ ($\Delta - 20$) $\implies$ Pole $\frac{1}{1+s/20}$. At $\omega = 200$, slope $0 \to -40$ ($\Delta - 40$) $\implies$ Double Pole $\frac{1}{(1+s/200)^2}$.
$\implies H(s) = \frac{0.5s}{(1+s/20)(1+s/200)^2}$.

11. Sampling Theorem & Aliasing

Ideal Sampling: $x_s(t) = x(t) \cdot \text{comb}_{T_s}(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s)$.
Sampled Spectrum:

$$X_s(f) = f_s \sum_{k=-\infty}^{\infty} X(f - kf_s)$$

Spectrum consists of periodic copies of the original spectrum $X(f)$, scaled by f_s, and shifted by integer multiples of the sampling frequency f_s.



Nyquist Rate (f_{Nyq}): Minimum sampling frequency to avoid aliasing. $f_{Nyq} = 2f_{max}$.
Nyquist Interval: $T_{max} = \frac{1}{2f_{max}}$.
Spectrum Gaps (Guard Band): Gap between baseband and adjacent spectral copies: Gap $= f_s - 2f_{max}$.
Aliasing Calculations: If sampled below Nyquist ($f_s < 2f_{max}$), images overlap. A frequency f_{in} folds into the baseband $[-f_s/2, f_s/2]$ at the alias frequency:

$$f_{alias} = |f_{in} - kf_s|$$

where k is an integer chosen to bring result into baseband $[0, f_s/2]$.

12. Signal Reconstruction & Filtering

To perfectly recover $x(t)$, use an Ideal Low-Pass Filter (Reconstruction Filter) to isolate the baseband copy.
Filter Gain: $K = T_s = 1/f_s$. (To cancel the f_s scaling factor).
Cutoff Freq (f_c): Must isolate the baseband without catching the image.
Condition: $f_{max} \le f_c \le f_s - f_{max}$. Usually placed at $f_c = f_s/2$.
Filter Equation: $H_r(f) = T_s\,\text{rect}(\frac{f}{2f_c})$.
Time Domain Interpolation (Sinc):
$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\,\text{sinc}(2f_c(t - nT_s))$.
Anti-Aliasing Filter: An analog low-pass filter applied *prior* to sampling to strictly bandlimit the signal to $f_c \le f_s/2$, completely removing high frequencies so no aliasing occurs.
Flat-Top Sampling (Practical): A Zero-Order Hold (ZOH) circuit keeps the sample value constant for duration τ. This multiplies the spectrum by a sinc function (Aperture Effect), rolling off high frequencies.
To correct this, the reconstruction filter must be cascaded with an **Equalizer Filter**: $H_{eq}(f) = \frac{1}{H_{ZOH}(f)} = \frac{\pi f\tau}{\sin(\pi f\tau)}$.
Energy of Filtered Sampled Signals:
If $x_s(t)$ passes through filter $H(f)$, apply Rayleigh's Theorem: $E_{out} = \int_{-\infty}^{\infty}|H(f)|^2|X_s(f)|^2 df$.
If $H(f)$ is an ideal rect, restrict integration bounds to passband, square amplitude within that band.

13. Complex Math & Partial Fractions

Euler's Formulas: $e^{\pm j\theta} = \cos(\theta) \pm j\sin(\theta)$.
$\cos(\theta) = \frac{1}{2}\left(e^{j\theta} + e^{-j\theta}\right)$, $\sin(\theta) = \frac{1}{j2}\left(e^{j\theta} - e^{-j\theta}\right)$.
Roots of Complex Numbers: To find n-th roots of $z = re^{j\theta}$:

$$z^{1/n} = r^{1/n}e^{j\left(\frac{\theta + 360^\circ \cdot k}{n}\right)} \quad \text{for } k = 0, 1, \ldots, n-1$$

Partial Fraction Expansion (Heaviside Cover-up):
For $Y(s) = \frac{N(s)}{(s-p_1)(s-p_2)}$, coefficient for $\frac{A}{s-p_1}$ is found by evaluating $(s - p_1)Y(s)$ at $s = p_1$.
Repeated Poles: $Y(s) = \frac{B}{(s-p_1)^2} + \frac{C}{s-p_1}$.
Find B by covering up $(s - p_1)^2$ and evaluating at p_1.
Find C by differentiating the covered-up expression wrt s and evaluating at p_1.
Complex Conjugate Poles: If $Y(s)$ has roots $s = -\alpha \pm j\beta$, it forms $\frac{A}{s+\alpha-j\beta} + \frac{A^*}{s+\alpha+j\beta}$. Find $A = |A|e^{j\theta}$ using cover-up. The inverse Laplace translates to $2|A|e^{-\alpha t}\cos(\beta t + \theta)u(t)$.

14. Key Transform Properties (FT & LT)

Time Shifting: $x(t - t_0) \rightleftharpoons e^{-j2\pi f t_0}X(f)$ (LT: $e^{-st_0}X(s)$)
Frequency Shifting: $e^{j2\pi f_0 t}x(t) \rightleftharpoons X(f - f_0)$ (LT: $X(s - s_0)$)
Time Scaling: $x(at) \rightleftharpoons \frac{1}{|a|}X(\frac{f}{a})$ (LT: $\frac{1}{|a|}X(\frac{s}{a})$)
Time Differentiation: $\frac{dx(t)}{dt} \rightleftharpoons j2\pi f X(f)$ (LT: $sX(s) - x(0^-)$)
Time Integration: $\int_{-\infty}^{t} x(\tau)d\tau \rightleftharpoons \frac{X(f)}{j2\pi f} + \frac{X(0)}{2}\delta(f)$ (LT: $\frac{X(s)}{s}$)
Modulation (Cosine): $x(t)\cos(2\pi f_c t) \rightleftharpoons \frac{1}{2}[X(f - f_c) + X(f + f_c)]$
Convolution in Time: $x_1(t) * x_2(t) \rightleftharpoons X_1(f) \cdot X_2(f)$ (LT: $X_1(s) \cdot X_2(s)$)

Found: https://github.com/brianstm/NUS.git